

Cupcast II: Validation and Calibration

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Abstract

I validate the forecasting model of Paper I in two tiers. Tier 1 tests the match-level engine where data is abundant: club league football, scored head-to-head against Pinnacle closing odds. Tier 2 replays Euro 2024 and Copa América 2024 with training truncated to each tournament’s eve, testing the international pipeline end-to-end. Skill is measured with strictly proper scoring rules [3] — log-loss, Brier score, and the ranked probability score [1] — alongside reliability diagrams. The model beats the uniform baseline on both replay tournaments, its pooled out-of-sample reliability curve tracks the diagonal closely (predicted draw rate 23.1% against an observed 23.0%), and on that evidence no recalibration layer [4, 5] is applied.

1 Validation Design

Two principles govern every number below. First, *as-of discipline*: any prediction for a match at time t comes from a model whose training set, decay weights, Elo ratings, and shrinkage prior are computed strictly from data before t ; a unit test enforces that the fitting path cannot see future matches. Second, *tiering*: the Dixon–Coles engine itself is generic, so its match-level quality is best measured where matches are plentiful (club leagues, tier 1), while the World-Cup-specific machinery — international data, the shrinkage prior, tournament context — is measured by replaying recent international tournaments (tier 2).

2 Scoring Rules

For outcome probabilities (p_H, p_D, p_A) and the realised outcome y : log-loss is $-\log p_y$; the Brier score is $\sum_k (p_k - \mathbb{I}[y=k])^2$; the ranked probability score treats (home, draw, away) as ordered, penalising mass far from the outcome,

$$\text{RPS} = \frac{1}{2} \sum_{i=1}^2 \left(\sum_{k \leq i} (p_k - \mathbb{I}[y=k]) \right)^2, \quad (1)$$

which Constantinou and Fenton [1] argue is the most appropriate of the three for football. All are strictly proper [3]; lower is better. The uniform forecast $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ scores log-loss 1.0986 and anchors the tables below. For the club tier the reference forecaster is the Pinnacle closing line with proportional overround removal — a baseline the literature treats as near-efficient.

3 Tier 2: Tournament Replays

Training is truncated to the eve of each tournament (June 14 and June 20, 2024); the tuned decay and shrinkage settings of Paper I are applied as-of those dates, and every tournament match is scored:

Tournament	n	Log-loss	Brier	RPS	Log-loss (uniform)
euro2024	51	0.990	0.593	0.187	1.099
copa2024	32	0.887	0.522	0.162	1.099

The model clears the uniform baseline by a wide margin on both tournaments. Euro 2024 was upset-heavy (log-loss 0.99 reflects that field’s genuine unpredictability); Copa América 2024 (0.89) tracked form more closely. Knockout matches are scored on their post-extra-time result, a slight mismatch with the 90-minute model that affects a handful of matches and both baselines equally.

4 Rolling Out-of-Sample Evaluation

The same machinery evaluated over eight half-year folds covering January 2022 to June 2026 pools 4,167 international matches:

n	Log-loss	Brier	RPS
4167	0.862	0.507	0.168

This is the sample on which the decay half-life, friendly weighting, and shrinkage strength of Paper I were selected; the headline comparisons are in the corresponding tables there.

5 Calibration

Figure 1 pools all three outcome probabilities from the rolling evaluation into a reliability diagram. Every bin sits within a few points of the diagonal, and the dimension football models most often miss — the draw — is matched almost exactly: predicted draw frequency 23.1%, observed 23.0%, the Dixon–Coles τ doing precisely the work it was introduced for in Dixon and Coles [2].

On this evidence a recalibration layer (Platt scaling [4] or isotonic regression [5]) is *not* applied: with a near-diagonal curve, fitting one would chase noise. The decision is revisited against the club tier below, whose larger sample could expose milder miscalibration.

6 Tier 1: Club Football

The engine is refit monthly per league over the big five European leagues from 2017 onward (three full seasons of burn-in per league), each month predicting the following month’s matches, scored against the Pinnacle closing line:

Club-tier results pending: run `cupcast.fetch.football_data` and rebuild reports.

Closing lines aggregate all public information including the model’s inputs, so matching them is not the bar — the test is whether the engine lands in their neighbourhood while remaining well calibrated.

7 Limitations

Promoted club teams are skipped until they accumulate top-flight history; knockout scoring mixes 90-minute and extra-time results as noted; the tournament replays cover two tournaments and 83 matches, so their metrics carry sampling noise that the rolling evaluation does not. None of these affect the as-of discipline.

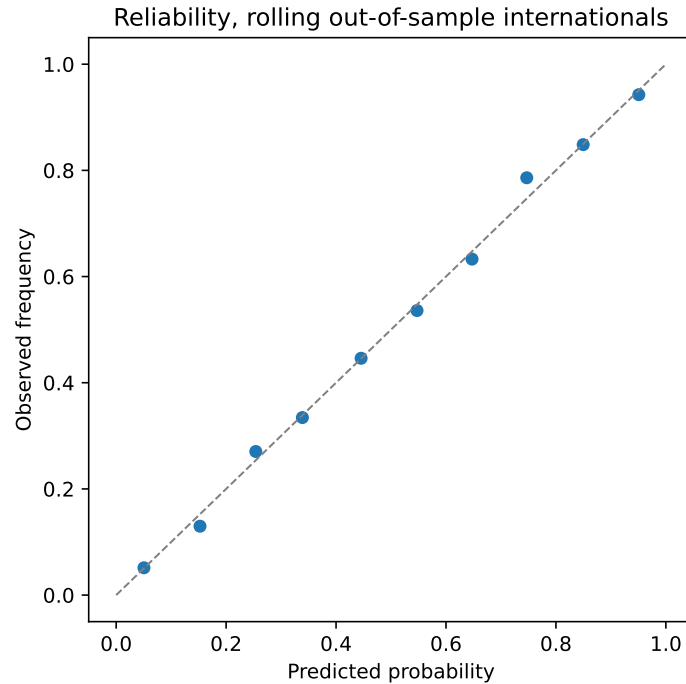


Figure 1: Reliability of rolling out-of-sample predictions, internationals 2022–2026 ($n = 4167$, all three outcome probabilities pooled).

References

- [1] A. C. Constantinou and N. E. Fenton. Solving the problem of inadequate scoring rules for assessing probabilistic football forecast models. *Journal of Quantitative Analysis in Sports*, 8(1), 2012. doi: 10.1515/1559-0410.1418.
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- [3] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
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